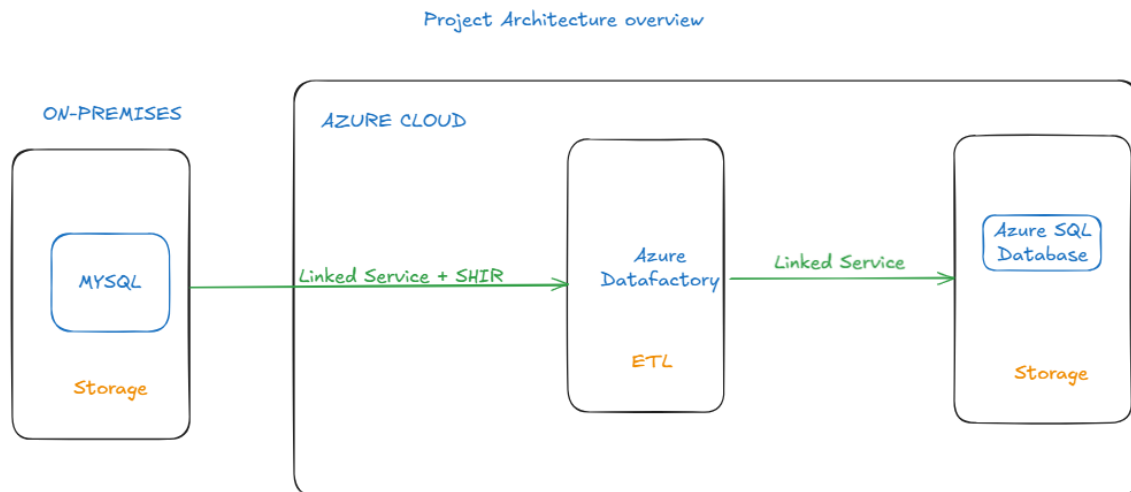


PROJECT OBJECT:

To Migrate structured data from an **on-premises MySQL database** to **Azure SQL Database** using a secure, dynamic, and reusable pipeline architecture built with **Azure Data Factory (ADF)**.

PROJECT ARCHITECTURE:



RESOURCES REQUIRED:

- [MYSQL Work Bench](#)
- **Azure Data Factory (ADF)** – Core orchestration tool.
- **Self-hosted Integration Runtime (IR)** – Secure connection to on-premise SQL Server.
- **Azure SQL Database** – Target cloud data store

CREATE A RESOURCE GROUP:

Azure services



Resources

Recent Favorite

Name	Type	Last Viewed
PRACTICE_RG	Resource group	a few seconds ago
Azure subscription 1	Subscription	16 hours ago

CREATE ADF:

Create Data Factory ...

Basics Git configuration Networking Advanced Tags Review + create

One-click to create data factory with sample pipeline and datasets. [Try it](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ

Azure subscription 1

Resource group * ⓘ

PRACTICE_RG

[Create new](#)

Instance details

Name * ⓘ

onprem2sqladf ✓

Region * ⓘ

Australia Southeast

[Previous](#)

[Next](#)

[Review + create](#)

CREATING THE IR:

Integration runtime setup

Network environment:

Choose the network environment of the data source / destination or external compute to which the integration runtime will connect to for data flows, data movement or dispatch activities:



Azure

Use this for running data flows, data movement, external and pipeline activities in a fully managed, serverless compute in Azure.



Self-Hosted

Use this for running data movement, external and pipeline activities in an on-premises / private network by installing the integration runtime.

Note: Data flows are only supported on Azure integration runtime. You can use self-hosted integration runtime to stage the data on cloud storage and then use data flows to transform it.

[View less](#) ^

External Resources:

You can use an existing self-hosted integration runtime that exists in another resource. This way

[Continue](#)

[Cancel](#)

Integration runtime setup

Private network support is realized by installing integration runtime to machines in the same on-premises network/VNET as the resource the integration runtime is connecting to. Follow below steps to register and install integration runtime on your self-hosted machines.

Name * ⓘ

ⓘ Validating the integration runtime name...

Description

Type

Self-Hosted

Create

Back

Cancel

Option 1: Express setup

[Click here to launch the express setup for this computer](#)

Option 2: Manual setup

Step 1: **Download and install integration runtime**

Step 2: Use this key to register your integration runtime

Name

Authentication key

Key1

IR@c260a764-b5b1-4019-966d-31273206756b@onprem2sqladf@Se



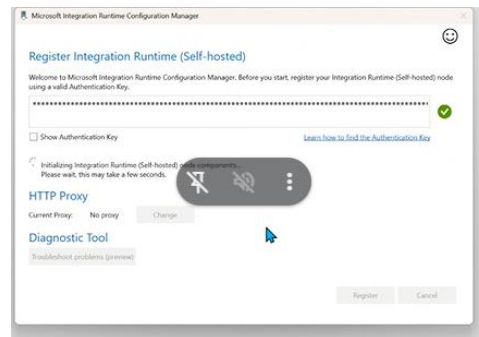
Key2

IR@c260a764-b5b1-4019-966d-31273206756b@onprem2sqladf@Se



Close

INSTALL MICROSOFT INTEGRATION RUNTIME:



New linked service

MySQL [Learn more](#)

Name *

MySQL1

Description

Connect via integration runtime *

selfhostintegrationRuntime1

Integration runtime selfhostintegrationRuntime1 is currently unavailable. Please try creating linked service once it is online.

Server name *

Port

3306

Database name *

User name *

Password

Azure Key Vault

Password *

SSL mode

Preferred

☐ Use system trust store ☒ Not use system trust store

Additional connection properties

+ New

Annotations

+ New

> Parameters

> Advanced

New linked service

 MySQL [Learn more](#) 

Name *

MySql1

Description

Connect via integration runtime * 



selfhostintegrationRuntime1



Server name *

Port

3306

Database name *

User name *

Password

Azure Key Vault

Password *

SSL mode

Preferred



☐ Use system trust store ☒ Not use system trust store

Additional connection properties

 New

Annotations

 New

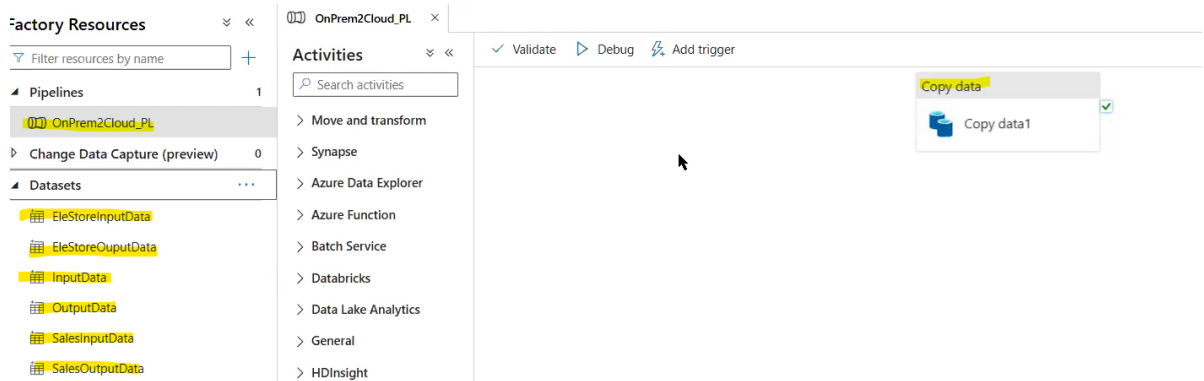
> Parameters

> Advanced 

USECASE1

- To Copy 1 Table Or To Copy Tables One By One.
- For Copy 3 Tables We Need 6 Datasets.(ele_store,bike,sales)

Example:



New linked service
MySQL [Learn more](#)

Name *
mysql2adf_ls

Description

Connect via integration runtime *
SelfHostedIntegrationRuntime

⚠ The credentials are stored in the machines of self-hosted integration runtime if you don't choose to store them in Azure Key Vault.

Server name *
localhost

Port
3306

Database name *
amazon_db

User name *
root

Password **Azure Key Vault**

Password *

SSL mode

Create **Cancel**

✓ Connection successful
[Test connection](#)

DATASET CREATION:

SOURCE:

Set properties

Name
InputData

Linked service *
mysql2adf_ls

Connect via integration runtime * ⓘ
Self-HostedIntegrationRuntime

Table name
Select...
Filter...
bang_data ('bang_data')
car_data ('car_data')
chen_data ('chen_data')
cust ('cust')
customer_info ('customer_info')
customer_info_1 ('customer_info_1')
customer_tb ('customer_tb')
ele_store ('ele_store')

OK Back Cancel

SINK:

General Source **Sink** Mapping Settings User properties

Sink dataset *
OutputData Open + New Learn more ⓘ

Write behavior
☒ Insert ☐ Upsert ☐ Stored procedure

Bulk insert table lock ⓘ
☐ Yes ☒ No

Table option
☐ Use existing ☒ Auto create table ⓘ
Add dynamic content [Alt+Shift+D]

Pre-copy script ⓘ



Azure SQL Database
OutputData

Connection Schema Parameters

Linked service *

adf2asql_js

Test connection Edit + New Learn more

Table

schema name

catData_Tbl

Preview data

Enter manually

Add dynamic content (Alt+Shift+D)

All pipeline runs > pipeline1 - Activity runs

Run Cancel Refresh Update pipeline Save Send

Copy data
Copy data1

Activity runs

Pipeline run ID 789790ba-d6d4-4318-b0d0-34962b7776d

All status

Showing 1 - 1 items

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties	Activity run ID	Log
Copy data1	Succeeded	Copy data	7/26/2015, 3:48:53 PM	2m 32s	selfhostedintegrationruntime1		c36795e-143c-4472-b4c9-1e688f44b0dc	Log

Monitor in Azure Metrics Export to CSV

Query 1 Query 2

Run Cancel query Save query Export data as Show only Editor

Shift + Enter

```
count(*) FROM [dbo].[catdataTbl]
```

Results Messages

Search to filter items...

301

USECASE:II

PROBLEM:

- With use case -1 is to migrate 50 tables , we need to have 100 datasets.
- Data Migration using Dataset Parameters.

WE NEED:

- 1 pipeline
- 2 datasets(but during the pipeline run time we need to give table name manually)
- **IMPLEMENTATION STEPS:**

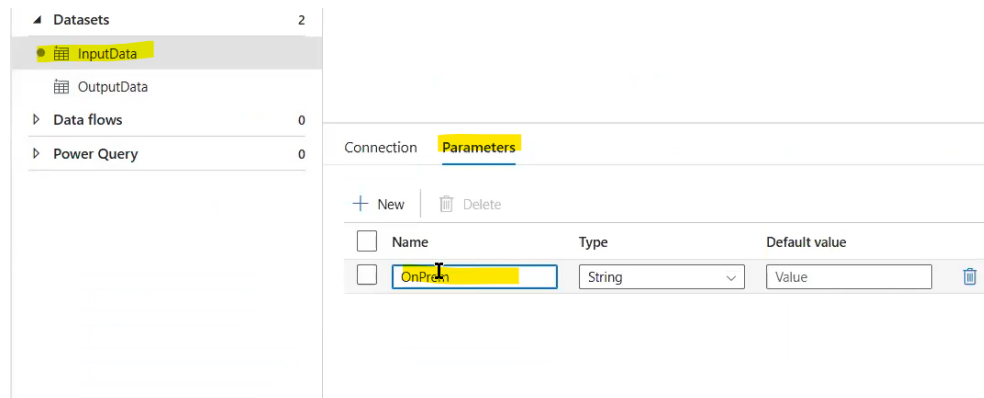
STEP1: CREATE NEW PIPELINE: COPY ACTIVITY

- **SOURCE:**
 - **Dataset:**
 - Inputdata
 - **Linkedservice:**
 - Mysql2adf_ls
 - **TableName**
 - car_data

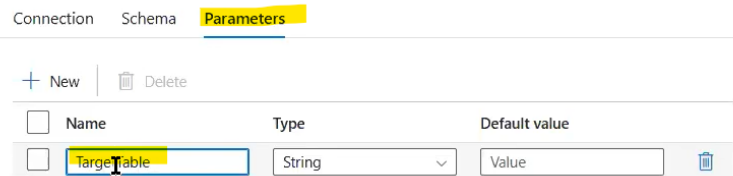
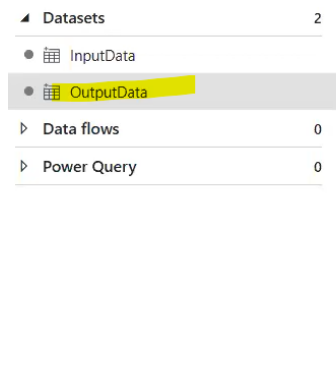
- **SINK:**
 - Dataset:
 - Outputdata
 - Linkedservice:
 - Adf2sql_ls
 - TableName:
 - Don't give any table name now.
 - Now goto *Table Option* : **select** Auto create table
 - Now open the **outputdata** --→ Enable the **Enter Manually**-→ Give the table name as: **Car_tbl**.(but this is same as usecase1)
 - So we need to implement the parameters in the source dataset and sink dataset.

STEP2: CREATING PARAMETERS:

- **Input Data:**
 - Parameters:
 - Name: OnpremTableName

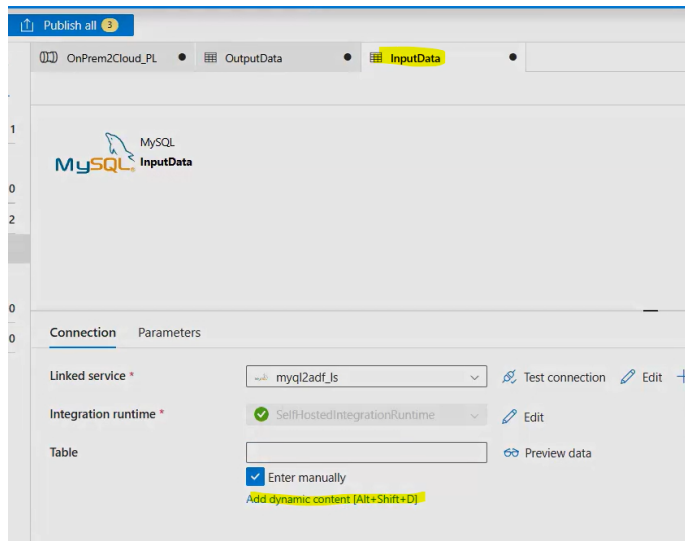


- **Output Data:**
 - Parameters:
 - Name: TargetTableName

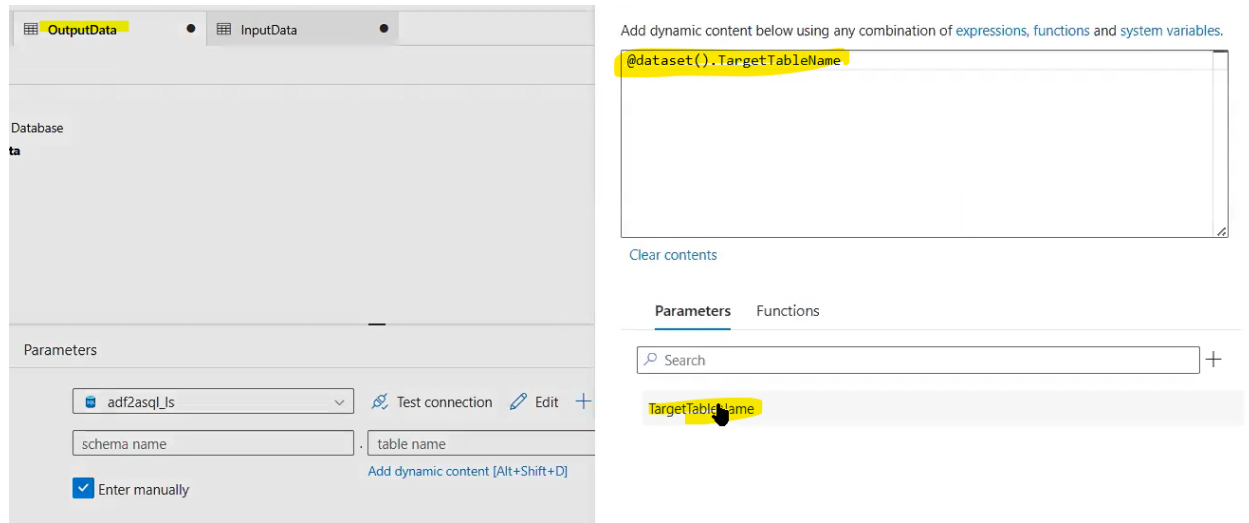


STEP3: UPDATING THE PIPELINE

- **Input Data:**
 - Remove the Manually given car_data.
 - Add Dynamic content.

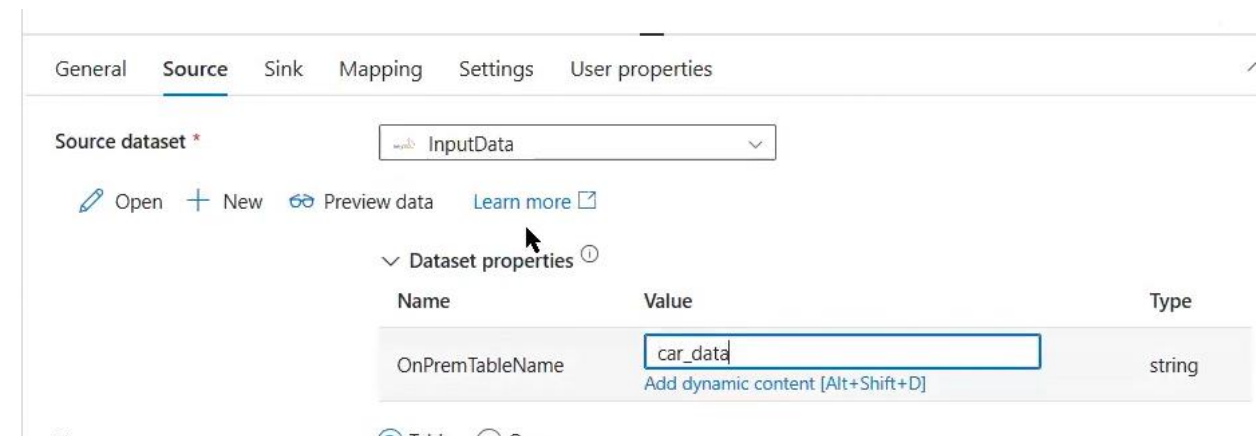


- **Output Data:**
 - Remove the Manually given car_tbl.
 - Add Dynamic content.



STEP 4: WHILE RUNNING THE PIPELINE

- 1. Move Car_Data:
 - Now we need to move the table.
- **Source:**



Sink:

General Source **Sink** Mapping Settings User properties

Sink dataset * [Open](#) [+ New](#) [Learn more](#)

Dataset properties ⓘ

Name	Value	Type
TargetTableName	Car_Tb Add dynamic content [Alt+Shift+D]	string

Write behavior ☒ Insert ☐ Upsert ☐ Stored procedure

Output:

Activity runs

Pipeline run ID f4e16744-107a-4508-989b-c706067bb735

All status

[Monitor in Azure Metrics](#) [Export to CSV](#)

Showing 1 - 1 items

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime	User prop...	Activity run ID		
Copy data1				Succeeded	Copy data	7/21/2025, 7:21:48 AM	38s	Self-HostedIntegrationRuntime	b71fb84a-5f16-4577-a0ac-fd3c7486d064

○ 2. Move Electronic_Data:

Source:

General **Source** Sink Mapping Settings User properties

Source dataset * [Open](#) [+ New](#) [Preview data](#) [Learn more](#)

Dataset properties ⓘ

Name	Value
OnPremTableName	ele_store Add dynamic content [Alt+Shift+D]

Use query ☒ Table ☐ Query

Additional columns ⓘ [+ New](#)

Sink:

General	Source	Sink	Mapping	Settings	User properties
---------	--------	-------------	---------	----------	-----------------

Sink dataset *

OutputData  Open  New [Learn more](#) 

Dataset properties 

Name	Value
TargetTableName	EleS(Tb Add dynamic content [Alt+Shift+D]

- 3. Move Sales_Data:

Source:

General	Source	Sink	Mapping	Settings	User properties
---------	---------------	------	---------	----------	-----------------

Source dataset *

web: InputData  Open  New  Preview data [Learn more](#) 

Dataset properties 

Name	Value
OnPremTableName	product_tb Add dynamic content [Alt+Shift+D]

Sink:

General	Source	Sink	Mapping	Settings	User properties
---------	--------	-------------	---------	----------	-----------------

Sink dataset *

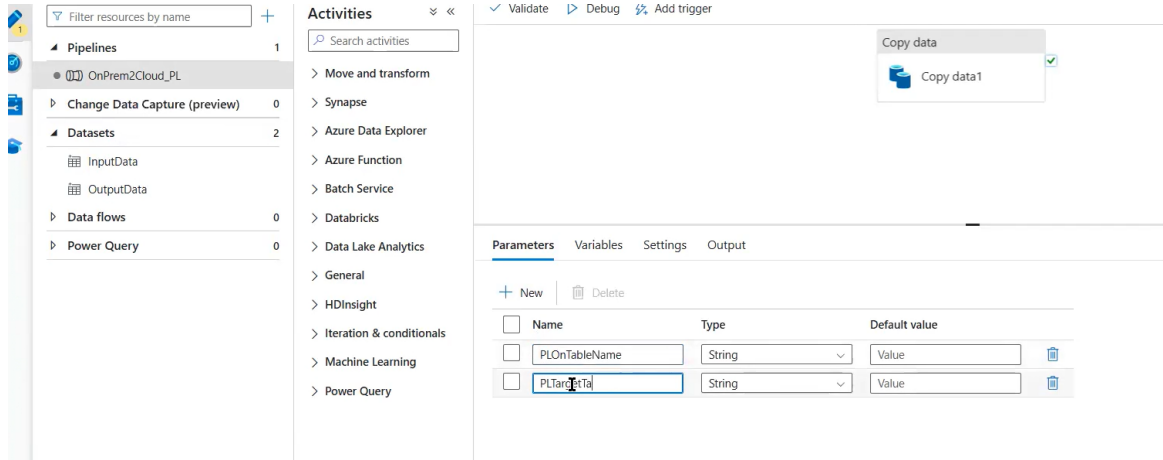
OutputData  Open  New [Learn more](#) 

Dataset properties 

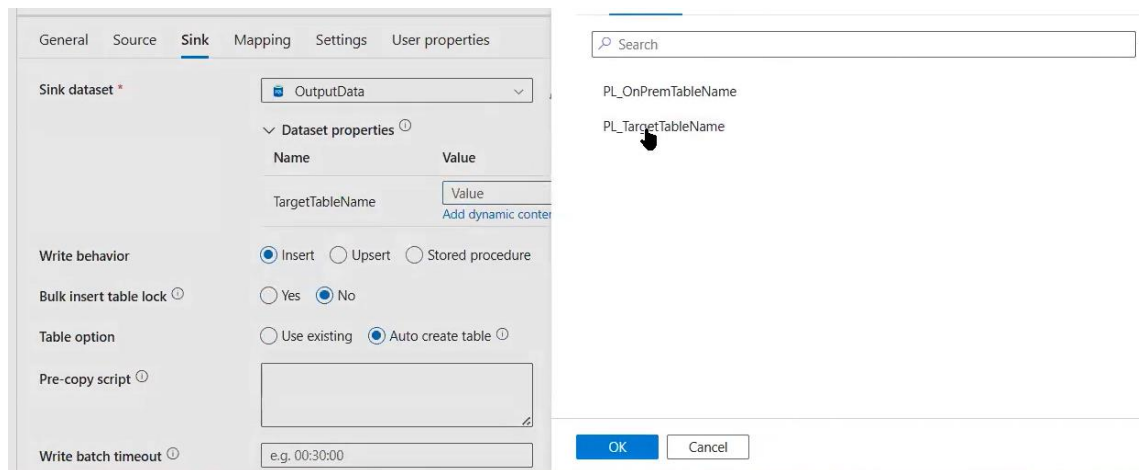
Name	Value
TargetTableName	Prod_Tb Add dynamic content [Alt+Shift+D]

USECASE3:

- We can create the parameter at the pipeline level.
- Along with the dataset parameters.



SOURCE:



SINK:

GeneralSourceSinkMappingSettingsUser properties

Sink dataset *

OutputData

Dataset properties

Name	Value
TargetTableName	Value Add dynamic content

Write behavior

☒ Insert

☐ Upsert

☐ Stored procedure

Bulk insert table lock

☐ Yes

☒ No

Table option

☐ Use existing

☒ Auto create table

Pre-copy script

Write batch timeout

e.g. 00:30:00

PL_OnPremTableName

PL_TargetTableName

OK

Cancel

Run the pipeline:

Pipeline run

Trigger pipeline now using last published configuration.

Parameters

Name	Type	Value
PL_OnPremTableName	string	Value
PL_TargetTableName	string	Value

OK

Cancel

Pipeline run

⚠️ Trigger pipeline now using last published configuration.

Parameters

Name	Type	Value
PL_OnPremTableName	string	student_marks_tb
PL_TargetTableName	string	Value

OK Cancel

USECASE-4

- If my table has 50 tables, 1 copy activity should run 50 times because of 50 tables.
- We need:
 - 1 copy activity.

Lets do it how:

- In mysql we have 22 tables. we need to copy all the tables in a single go.
- To see all the tables names in mysql:

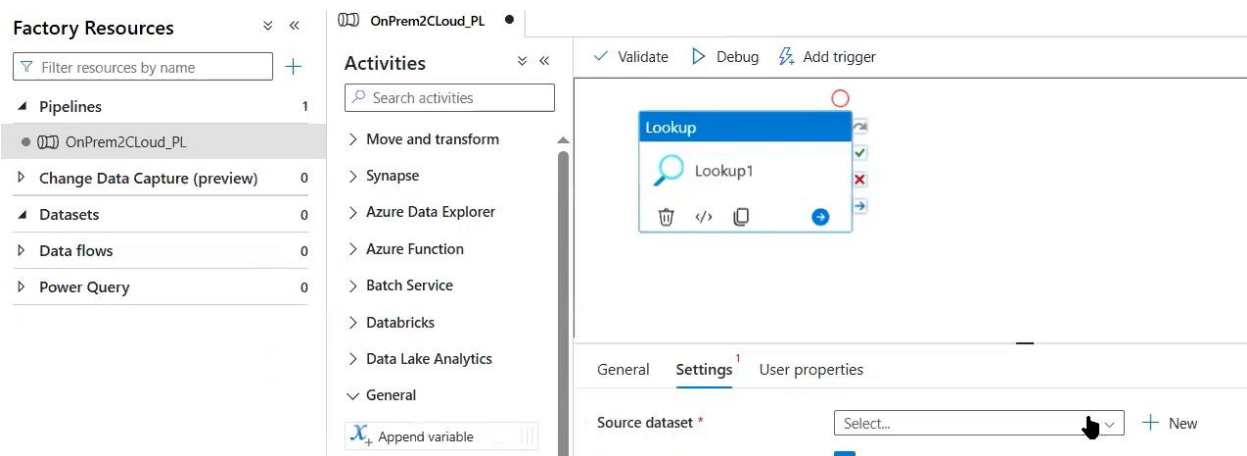
Select tablename from information_schema.tables

Where table_schema="amazon_db"

And table_type="base Table";

- To give table name dynamically instead of giving manually one by one.
- Follow the steps:

STEP1: ADD THE LOOKUP ACTIVITY



CREATE DATASET:

Set properties

Name

OnPremTabl

Linked service *

Select...

CREATE LS:

Set properties

Name

OnPremTables

Linked service *

mysql2adf_ls

Connect via integration runtime * ⓘ

Self-HostedIntegrationRuntime

Table name

Loading...

☐ Enter manually

> Advanced

WRITE A QUERY IN THE LOOKUP ACTIVITY:



General **Settings** ¹ User properties

Source dataset * OnPremTables

[Open](#) [+ New](#) [Preview data](#) [Learn more](#)

First row only ☒

Use query ☐ Table ☒ Query

Query *

```
SELECT table_name FROM  
information_schema.tables WHERE  
table_schema = 'amazon_db'  
and table_type = 'BASE TABLE';
```

[Edit](#)

[Add dynamic content \[Alt+Shift+D\]](#)

Output:

Activity runs

Pipeline run ID 7745255e-1489-4bbe-9fd3-467144badfc2

All status ▾

Showing 1 - 1 items

Activity name	Activity st...	Activit...	Run start	Duration
Lookup1	✓ Succeeded	Lookup	7/22/2025, 7:00:32 AM	15s

Output

Copy to clipboard

```
{
  "firstRow": {
    "TABLE_NAME": "car_d[ita"
  },
  "effectiveIntegrationRuntime":
"Self-HostedIntegrationRuntime",
  "billingReference": {
    "activityType": "PipelineActivity",
    "billableDuration": [
```

- We are getting the tablename of first table, but we need the names for all the tables.
- To do goto update pipelin-→disable the First row only checkbox:

General

Settings

User properties

Source dataset *

OnPremTables

Open

New

Preview data

Learn more

First row only

☒

Use query

☐ Table ☒ Query

Query *

SELECT table_name FROM
information_schema.tables WHERE
table_schema = 'amazon_db'

Edit

- We are getting the tablename of first table, but we need the names for all the tables.
- To do goto update pipelin-→disable the First row only checkbox:

General **Settings** ¹ User properties

Source dataset *  Open  New 

First row only ☐

Use query ☐ Table ☒ Query

Query *

`table schema = 'amazon_db'`
`and table type = 'BASE TABLE';`

 Edit

Add dynamic content [Alt+Shift+D]

[RUN THE PIPELINE NOW:](#)

Output: output of every activity will generate the JSON file as a output.

Table count : 22

Output  

 Copy to clipboard

```

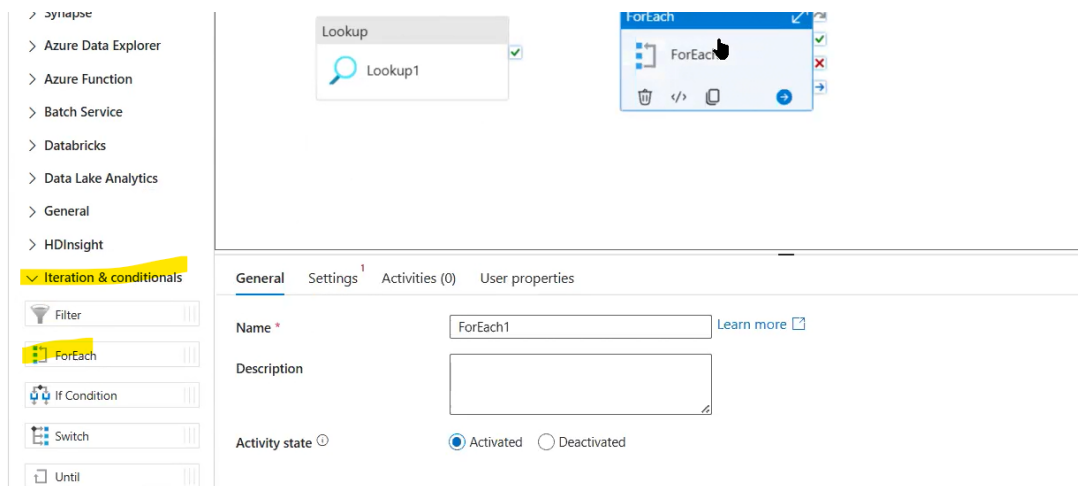
{
  "count": 22,
  "value": [
    {
      "TABLE_NAME": "car_data"
    },
    {
      "TABLE_NAME": "cust"
    }
  ]
}

```

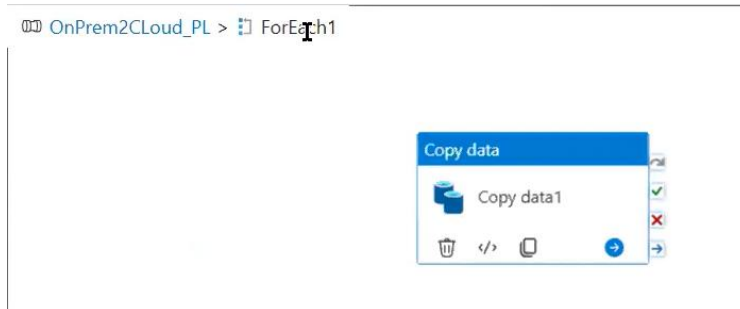
Step2: we need to pass all the tables name one by one to copy activity, for that we need for each activity.

If we are doing lookup to copy, only first table name will be passed. But we need all the tables name.

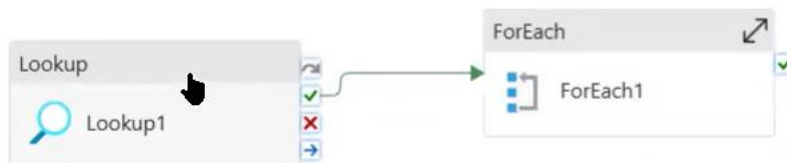
Iteration-→for each



Double click the for each now add the copy activity



Now connect lookup and for each



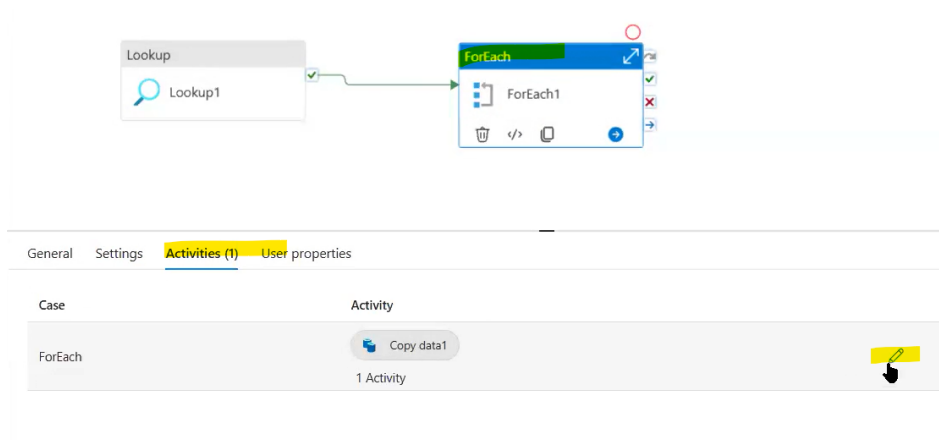
Need to send output of lookup to for each:

Lookup-→setting:

The screenshot displays the Visual Studio Code interface for configuring a workflow. At the top, a workflow diagram shows a 'Lookup' activity connected to a 'ForEach' loop. Below this, the 'Settings' tab for the 'Lookup' activity is open. The 'Sequential' checkbox is unchecked. The 'Batch count' field is empty. The 'Items' field is set to '@activity('Lookup1').output.value'. To the right, a panel titled 'Add dynamic content below using any combination of expressions, functions and system variables.' is open. It shows a search bar and a list of activity outputs under the 'Activity outputs' tab. The outputs listed are: 'Copy data1' (with a 'Copy' button and a tooltip 'Click to insert expression'), 'Lookup1' (Lookup1 activity output), 'Lookup1 count' (Count of the rows), and 'Lookup1 value array' (Array of row data). The 'OK' button is visible at the bottom of the panel.

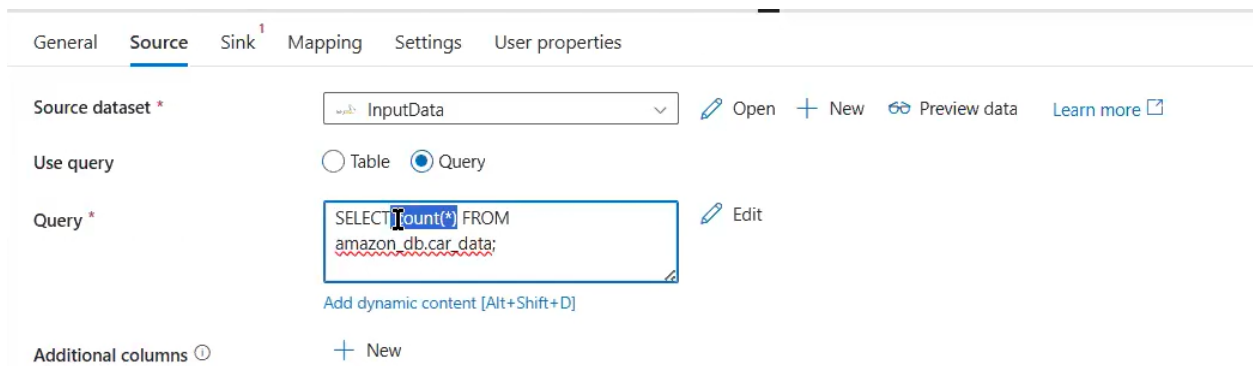
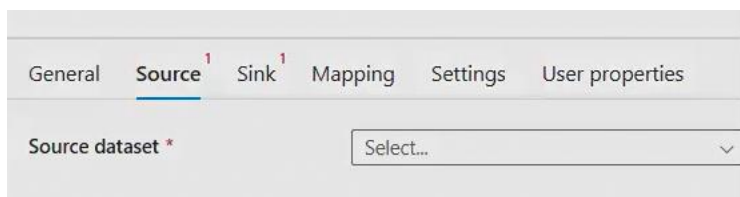
You can select any from the above option.

Inside the foreach we have copy activity.



Copy activity:

Source:



We will get the tablename from lookup activity.

OnPremTableList

✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

OnPrem2Cloud_PL > ForEach1

General **Source** Sink Mapping Settings User properties

Source dataset * InputData

Use query ☐ Table ☒ Query

Query *

Add dynamic content [Alt+Shift+D]

Additional columns ⓘ + New

Pipeline expression builder

Add dynamic content below using any combination of [expressions](#), [functions](#) and [system variables](#).

I

Clear contents

ForEach iterator

Activity outputs

Parameters

System variables

...

Search

ForEach1

Current item

@item().TABLE_NAME

So the query should be:

Pipeline expression builder

Add dynamic content below using any combination of [expressions](#), [functions](#) and [system variables](#).

```
@concat('SELECT * FROM ',item().TABLE_NAME)
```

Target: sink:

The screenshot shows the 'Sink' tab of the 'OnPrem2Cloud_PL' pipeline. The 'Sink dataset' dropdown is set to 'Select...'. To the right, a 'Select a data store' dialog is open, displaying a grid of data store options. The 'All' tab is selected, showing options like NoSQL, Azure Database for MySQL, Azure Database for PostgreSQL, Azure Databricks Delta Lake, Azure File Storage, Azure SQL Database, and Azure SQL Database Managed Instance.

In pipeline activities and data flows, reference a dataset to specify the location and structure of data within a data store. [Learn more](#)

Select a data store

Search

All	Azure	Database	File	Generic proto
NoSQL		(Kusto)		Gen2
Azure Database for MySQL	Azure Database for PostgreSQL	Azure Databricks Delta Lake		
Azure File Storage	Azure SQL Database	Azure SQL Database Managed Instance		

In the output data:

The screenshot shows the 'Output Data' tab of the pipeline. The 'Table' field is set to 'schema name'. The 'Enter manually' checkbox is checked. The 'Linked service' is set to 'adf2asql_Is'. The 'Table' field is also set to '@dataset().Tab_Name'.

Output Data

- Data flows 0
- Power Query 0

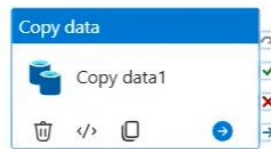
Connection Schema Parameters

Linked service * adf2asql_Is Test connection Edit + New Learn more

Table schema name Enter manually @dataset().Tab_Name Preview data

Sink:

OnPrem2Cloud_PL > ForEach1



General Source **Sink** Mapping Settings User properties

Sink dataset * OutputData Open New [Learn more](#)

Dataset properties

Name	Value
Tab_Name	Value Add dynamic content [Alt+Shift+D]

Write behavior ☒ Insert ☐ Upsert ☐ Stored procedure

Bulk insert table lock ☐ Yes ☒ No

OnPrem2Cloud_PL > ForEach1

Copy data

Copy data1

General Source **Sink** Mapping Settings User properties

Sink dataset * OutputData Open New [Learn more](#)

Dataset properties

Name	Value
Tab_Name	@item() 🗑️

Will fail.so need to give as follow.

General Source Sink Mapping Settings User properties

Sink dataset *

 OutputData

 Open  New [Learn more](#) 

Dataset properties ⓘ

Name

Value

Tab_Name

@item().TABLE_NAME

Write behavior

☒ Insert ☐ Upsert ☐ Stored procedure

Bulk insert table lock ⓘ

☐ Yes ☒ No